


Exam 2 review

Review Q11

random component:

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i} (1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

systematic component

$$\begin{cases} \log\left(\frac{\alpha_i}{1 - \alpha_i}\right) = \gamma_0 + \gamma_1 \text{Gender}_i + \gamma_2 \text{Age}_i + \gamma_3 \text{Income}_i + \\ \gamma_4 \text{Private}_i + \gamma_5 \text{Free Poor}_i + \gamma_6 \text{Free Repat}_i \\ \log(\lambda_i) = \beta_0 + \beta_1 \text{Gender}_i + \beta_2 \text{Age}_i + \beta_3 \text{Income}_i + \\ \beta_4 \text{Private}_i + \beta_5 \text{Free Poor}_i + \beta_6 \text{Free Repat}_i \end{cases}$$

Q 12 :

random component:

$$P(Y_i = y) = \begin{cases} e^{-\lambda_i} (1 - \alpha_i) + \alpha_i & y = 0 \\ \frac{e^{-\lambda_i} \lambda_i^y}{y!} (1 - \alpha_i) & y > 0 \end{cases}$$

went estimated # of doctor visits

Two options: $Z_i = 1$ (belong to "always 0" group)
 $P(Z_i = 1) = \alpha_i \Rightarrow Y_i = 0 \Rightarrow \text{predicted value (average value)} = 0$

$Z_i = 0$ (belong to Poisson group, "Sometimes 0")
 $P(Z_i = 0) = 1 - \alpha_i \Rightarrow Y_i | (Z_i = 0) \sim \text{Poisson}(\lambda_i)$
 $\Rightarrow \text{predicted value (average value)} = \lambda_i$

Expected # of visits = $0 \cdot \alpha_i + \lambda_i (1 - \alpha_i) = \lambda_i (1 - \alpha_i)$
estimate = $\hat{\lambda}_i (1 - \hat{\alpha}_i)$

$$\text{estimate} = \hat{\lambda}_i (1 - \hat{\alpha}_i)$$

$$\begin{aligned} \hat{\lambda}_i &= \exp \left\{ 0.12 + 0.37 \left(\frac{20}{100} \right) - 0.41 \left(\frac{5000}{10000} \right) \right\} \\ &= 0.99 \end{aligned}$$

$$\log \left(\frac{\hat{\alpha}_i}{1 - \hat{\alpha}_i} \right) = 1.7 - 0.8 \left(\frac{20}{100} \right) - 0.2 \left(\frac{5000}{10000} \right) = 1.44$$

$$\Rightarrow \hat{\alpha}_i = \frac{e^{1.44}}{1 + e^{1.44}} = 0.808$$

$$\Rightarrow 0.99 (1 - 0.808) = \boxed{0.19}$$

Quasi-Poisson models

Poisson:

$$Y_i \sim \text{Poisson}(\lambda_i) \rightarrow$$

$$\text{mean of } Y_i = \lambda_i$$

$$\text{var}(Y_i) = \lambda_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 X_i$$

Sometimes $\text{var}(Y_i) \gg \text{mean of } Y_i$ in observed data

$\Rightarrow$ Poisson is not a good fit

$$\phi = \frac{\text{var}(Y_i)}{\text{mean}(Y_i)}$$

(if Poisson is correct, $\phi = 1$)

if we have overdispersion, then $SE(\hat{\beta})_{\text{poisson}}$ are too small

$\Rightarrow$ CIs are too narrow,
p-values are too small

Quasi-Poisson model:

$$\text{mean of } Y_i = \lambda_i, \quad \text{var}(Y_i) = \phi \lambda_i$$

$$\log(\lambda_i) = \beta_0 + \beta_1 X_i$$

estimate ϕ , then

$$SE(\hat{\beta})_{QP} = \sqrt{\hat{\phi}} \cdot SE(\hat{\beta})_{\text{Poisson}}$$

$\hat{\beta}$ is the same, $SE(\hat{\beta})$ is bigger

estimating ϕ :

$$\hat{\phi} = \frac{1}{n-p} \sum_{i=1}^n e_{(n)i}^2$$

Pearson residuals : $e_{(n)i} = \frac{y_i - \hat{\lambda}_i}{\sqrt{\hat{\lambda}_i}} \leftarrow \text{raw residual } y_i - \hat{\lambda}_i \text{ (observed - predicted)}$
 $\leftarrow \text{std. dev. if Poisson is correct}$

if Poisson is correct, $\text{Var} \left(\frac{y_i - \hat{\lambda}_i}{\sqrt{\hat{\lambda}_i}} \right) \approx 1$

if Poisson is wrong : $\sqrt{\hat{\lambda}_i}$ is not std dev of y_i ;
(correct std dev for GP = $\sqrt{\hat{\phi} \hat{\lambda}_i}$)

$\Rightarrow e_{(n)i}$ are bigger than 1

$\Rightarrow \hat{\phi}$ is bigger than 1

Likelihood ratio test

compare two nested models (full model - H_A
reduced model - H_0)

$$2 \log \left(\frac{L_{\text{full}}}{L_{\text{reduced}}} \right) = 2 \log L_{\text{full}} - 2 \log L_{\text{reduced}}$$

For Poisson = deviance_{reduced} - deviance_{full} (drop-in-deviance)
(or logistic)

For NB or ZIP: calculate log-likelihoods directly rather than computing deviance

For GP: nested F test (wait put this on exam)

GOF test

- defined specifically for Poisson model
- check whether Poisson model is good fit to data

H_0 : model is a good fit

H_A : model is not a good fit

Test Stat = residual deviance

Compare to χ^2_{n-p}